

Configure jack

Jack is the currently used Java toolchain for building LineageOS 14.1 and up. It is known to run out of memory often if not configured correctly - a simple fix is to run this command:

```
• export ANDROID_JACK_VM_ARGS="-Dfile.encoding=UTF-8 -XX:+TieredCompilation -Xmx4G"
```

NOTE

If compression is enabled, the cache size can be lower (aim for approximately 20GB for one device).

Adding that command to your `~/.bashrc` file will automatically configure Jack to allocate a sufficient amount of memory (in this case, 4GB).

Start the build

Time to start building! Now, type:

- `croot`
- `brunch oneplus3`

The build should begin.

Install the build

Assuming the build completed without errors (it will be obvious when it finishes), type the following in the terminal window the build ran in:

- `cd $OUT`

There you'll find all the files that were created. The two files of more interest are:

1. `recovery.img`, which is the LineageOS recovery image.
2. `lineage-15.1-20180520-UNOFFICIAL-oneplus3.zip`, which is the LineageOS installer package.

Success! So... what's next?

That's excellent. You are now part of a select few individuals from across the world who have managed to build their own Android OS from scratch. We hope you now start taking up new projects which involve making new